

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-305-MJ-P
Operating System-I

Duration: 3 Hours**Maximum Marks:35**

Q.1) Write a C Menu driven Program to implement following functionality

- a) Accept Available
- b) Display Allocation, Max
- c) Display the contents of need matrix
- d) Display Available

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	2	3	2	9	7	5	3	3	2
P1	4	0	0	5	2	2			
P2	5	0	4	1	0	4			
P3	4	3	3	4	4	4			
P4	2	2	4	6	5	5			

[15 Marks]

Q.2 Write a C program that behaves like a shell which displays the command

prompt '\$'. It accepts the command, tokenize the command line and execute it by creating the child process. Also implement the additional command 'count' as

- a. \$ count c filename: It will display the number of characters in given file
- b. \$ count w filename: It will display the number of words in given file
- c. \$ count l filename: It will display the number of lines in given file

[15 Marks]

Q.3. Oral/Viva

[5 Marks]